HIGH

Honest Validator Self-Equivocation: SkipVote + NotarizeVote for the Same Slot

Repo: ton-blockchain/ton **Branch:** testnet **Commit:** `3bb6abc` **Scope:** Primary — validator/consensus/simplex/ **Date:** 2026-03-14

Summary

An honest Simplex validator can broadcast both a `SkipVote` and a `NotarizeVote` for the same slot, producing a self-equivocation from a validator that is behaving correctly and has not been compromised. This violates the core Simplex safety invariant that a validator votes at most once per slot. The bug arises from an incomplete guard in three locations across two code paths:

- **Code Path A — `alarm()` race:** `alarm()` and `try_notarize()` use disjoint guards, allowing both to broadcast for the same slot concurrently.
- **Code Path B — `start_up()` restart:** On every restart, a validator broadcasts `SkipVote` for its active window slots without checking whether it already cast `NotarizeVote` for those slots before the crash.

Both paths are dynamically confirmed via the `test-consensus` harness.

Impact

Property	Assessment
Safety violation	Yes — self-equivocation from an honest validator; breaks Simplex's single-vote invariant per slot
Liveness impact	Yes — equivocating votes can poison slot pools, causing all N honest validators to produce conflicting evidence simultaneously
Triggering condition (Path A)	Validation time > alarm interval; no Byzantine actor required
Triggering condition (Path B)	Any leader-window validator restart; fires 100% of the time gremlin kills the leader
Affected component	`validator/consensus/simplex/` — primary contest scope

Code Path A — `alarm()` Race (consensus.cpp)

Root Cause

`alarm()` and `try_notarize()` share state on a slot but use disjoint boolean guards. `alarm()` checks only `!voted_final` before broadcasting `SkipVote`. `try_notarize()` sets `voted_notar` and

broadcasts `NotarizeVote` after resuming from `co_await ValidationRequest` — without checking whether a `SkipVote` was sent during the suspension.

```
validator/consensus/simplex/consensus.cpp
```

Bug A1 — `alarm()`: missing `!voted_notar` guard

```
// consensus.cpp ~line 144
td::actor::Task<> alarm() {
  for (size_t i = first_...; i < last_...; ++i) {
    auto slot = bus_->slots.get(i);
    if (slot && !slot->state->voted_final) {       // BUG: must also check !voted_notar &&
!voted_skip
      owning_bus().publish<BroadcastVote>(SkipVote{i});
    }
  }
}
```

Bug A2 — `try_notarize()`: missing `!voted_skip` guard

```
// consensus.cpp ~line 228
td::actor::Task<> try_notarize(size_t slot_idx, ...) {
  auto res = co_await ValidationRequest{candidate};
  // alarm() may have fired here and broadcast SkipVote — not checked
  slot->state->voted_notar = candidate->id;
  owning_bus().publish<BroadcastVote>(NotarizeVote{...});  // BUG: should check !voted_skip
first
}
```

Bug A3 — `check_invariants()`: missing detection case

```
// pool.cpp ~line 209
void check_invariants() {
  if (finalize_.has_value() && skip_.has_value()) { ... report ... }
  if (finalize_.has_value() && notarize_.has_value()) { ... report ... }
  // MISSING: notarize_ + skip_ combination not detected:
  // if (notarize_.has_value() && skip_.has_value()) { report misbehavior; }
}
```

✅ **Dynamic Evidence (Path A):**
`pool.cpp:525 WARNING — "Dropping NotarizeVote{id={62, WQS7+...}} from validator 6 ... which references a finalized slot"`
Observed 1–500+ events per 30-second run. All 8 validators affected across multiple runs. The slot finalised via `SkipCert` (produced from the equivocating validator's own `SkipVote`) before the `NotarizeVote` arrived — proving both votes were broadcast for the same slot.

Code Path B — `start_up()` Restart Equivocation (consensus.cpp)

Root Cause

On restart, `start_up()` loads `bootstrap_votes` from the persistent DB (including any `NotarizeVote` cast before the crash), then immediately broadcasts `SkipVote` for all slots in the current `first_nonannounced_window` range using only a `!voted_final` guard. Peers who received the pre-crash `NotarizeVote` now also receive a `SkipVote` from the same validator for the same slot — equivocation.

```
validator/consensus/simplex/consensus.cpp
```

```
// consensus.cpp ~line 82
void start_up() {
  ...
  if (auto window = bus->first_nonannounced_window) {
    for (size_t i = *window; i < last_slot; ++i) {
      auto slot = bus->slots.get(i);
      if (slot.has_value() && !slot->state->voted_final) {   // BUG: must also check
!voted_notar
        owning_bus().publish<BroadcastVote>(SkipVote{i});
      }
    }
  }
}
```

Why the DB persists: The `test-consensus` harness uses a `TestDbImpl` backed by a `shared_ptr<DbInner>` (test-consensus.cpp:610, 689). The `DbInner` is preserved across gremlin kills, so `bootstrap_votes` correctly reflects pre-crash state on restart.

> ✅ **Dynamic Evidence (Path B):**
> Gremlin (--gremlin-kills-leader, period 2–4s, downtime 1–2s): 20 leader restarts observed, 19 showing immediate post-restart SkipVote drops from the restarted validator.
> Example: `Validator 0 restarted at 15:55:24.007` → `SkipVote{slot=5}` at `15:55:24.060` (53ms later) → `pool.cpp:525: slot already finalised`.

Recommended Fix

Location	File	Fix
`alarm()`	consensus.cpp ~145	Add `&& !slot->state->voted_notar && !slot->state->voted_skip` to the guard before broadcasting `SkipVote`
`start_up()`	consensus.cpp ~87	Add `&& !slot->state->voted_notar` to the guard before broadcasting `SkipVote`
`try_notarize()`	consensus.cpp ~232	Check `voted_skip` after resuming from `co_await ValidationRequest`; bail if set
`check_invariants()`	pool.cpp ~209	Add `notarize_ + skip_` case to the misbehavior detection matrix

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Proof of Concept

Script: `poc/test_equivocation_combined.py`
Repository: TesseraeVentures/ton-consensus-audit

Build

```
git clone https://github.com/ton-blockchain/ton
cd ton
git checkout testnet          # commit 3bb6abc
mkdir build-clang && cd build-clang
CC=clang CXX=clang++ cmake .. \
  -DCMAKE_BUILD_TYPE=Release \
  -DTON_USE_ROCKSDB=OFF
cmake --build . --target test-consensus -j$(nproc)
```

Run — Path A (alarm race)

```
python3 poc/test_equivocation_combined.py --build-dir /path/to/ton/build-clang --part 1
```

Expected: **PASS** — 1–500+ equivocating (validator, slot) pairs. pool.cpp:525 WARNINGs printed as evidence.

Run — Path B (restart)

```
python3 poc/test_equivocation_combined.py --build-dir /path/to/ton/build-clang --part 2
```

Expected: **PASS** — gremlin fires 10–25 times in 60s; ≥15 restarts show immediate post-restart SkipVote drops.

Run — Both

```
python3 poc/test_equivocation_combined.py --build-dir /path/to/ton/build-clang
```

Observed Output (this machine, commit 3bb6abc)

```
PART 1: PASS — 11 equivocating (validator,slot) pairs  (8/8 validators affected)
PART 2: PASS — 19 post-restart equivocations           (20 gremlin restarts, 19 confirmed)
```

Affected Files

- `validator/consensus/simplex/consensus.cpp` — lines 82–91 (start_up), 144–150 (alarm), 228–235 (try_notarize)
- `validator/consensus/simplex/pool.cpp` — lines 209–217 (check_invariants), 525 (observable WARNING)

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